PARSHVANATH CHARITABLE TRUST'S

A. P. SHAH INSTITUTE OF TECHNOLOGY

Department of Information Technology

(NBA Accredited)

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Class / Branch: BE IT-B
Subject: Recent Open Source Lab
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EXPERIMENT NO. 05

Aim: To study open-source database management tools like Cassandra, MySql, MariaDB and understand the steps for contributing in Cassandra.

```
apsit@apsit-HP-280-G3-MT:~$ docker --version
Docker version 29.6.1, build 8900f1d
apsit@apsit-HP-280-G3-MT:~$
docker run --name cassandra-1 -d cassandra:latest
Unable to find image 'cassandra:latest' locally
latest: Pulling from library/cassandra
eaa1cb70b9f9: Pull complete
ee9bf2b77e96: Pull complete
02bc933523c1: Pull complete
4552b21f2e30: Pull complete
597c6c618d36: Pull complete
cc671df7c89f: Pull complete
07f9b4e253d0: Pull complete
67583a9db8ca: Pull complete
239d0ec75d63: Download complete
69179d32e7c9: Download complete
Digest: sha256:e52bb93c21f69cc5c2f5eb9e7d9736b10a047ad38bc7776480c7dbbf7d4e0ba7
Status: Downloaded newer image for cassandra:latest
77bb3eec57bed7f1e4f6db9f01f985a75a461750917757155cbca625f897cdc5
apsit@apsit-HP-280-G3-MT:~$ docker exec -it cassandra-1 nodetool status
Exception in thread "main" java.lang.IllegalArgumentException: Server is not initialized yet, cannot run nodetool.
        at org.apache.cassandra.tools.NodeTool.err(NodeTool.java:328)
        at org.apache.cassandra.tools.NodeTool.execute(NodeTool.java:291)
        at org.apache.cassandra.tools.NodeTool.main(NodeTool.java:87)
apsit@apsit-HP-280-G3-MT:~$ docker exec -it cassandra-1 bash -c 'cqlsh'
Connection error: ('Unable to connect to any servers', {'127.0.0.1:9042': ConnectionRefusedError(111, "Tried connecting to [('127.0.0.1', 9042
)]. Last error: Connection refused")})
```

```
apsit@apsit-HP-280-G3-MT:~$ docker exec -it cassandra-1 bash -c 'cqlsh'
Connection error: ('Unable to connect to any servers', {'127.0.0.1:9042': ConnectionRefusedError(111, "Tried connecting to [('127.0.0.1', 9042
)]. Last error: Connection refused")})
apsit@apsit-HP-280-G3-MT:~$ sudo docker exec -it cassandra-1 nodetool status
[sudo] password for apsit:
Datacenter: datacenter1
=======================
Status=Up/Down
|/ State=Normal/Leaving/Joining/Moving
--  Address     Load        Tokens  Owns (effective)  Host ID                               Rack
UN  172.17.0.2  114.71 KiB  16      100.0%            6ba99ffc-13fe-4f5f-b54f-d221720cfb36  rack1
```

```
apsit@apsit-HP-280-G3-MT:~$ docker exec -it cassandra-1 bash -c 'cqlsh'

ATTENTION: All commands will be saved to history file: /root/.cassandra/cqlsh_history
This may include sensitive information such as passwords.
To disable history, use --disable-history or set 'disabled = true' in the [history] section of cqlshrc.
See https://cassandra.apache.org/doc/latest/tools/cqlsh.html for more information.

Connected to Test Cluster at 127.0.0.1:9042
[cqlsh 6.2.0 | Cassandra 5.0.8 | CQL spec 3.4.7 | Native protocol v5]
Use HELP for help.
cqlsh> CREATE KEYSPACE labdemo
   ... WITH replication = {'class': 'SimpleStrategy', 'replication_factor' : 1};
cqlsh> USE labdemo;
cqlsh:labdemo> CREATE TABLE faculty (
           ...     facultyId int PRIMARY KEY,
           ...     name text,
Trash      ...     dept text
           ... );
cqlsh:labdemo> INSERT INTO faculty (facultyId, name, dept) VALUES (1, 'Clark', 'Comp');
cqlsh:labdemo> INSERT INTO faculty (facultyId, name, dept) VALUES (2, 'Dave', 'IT');
cqlsh:labdemo> INSERT INTO faculty (facultyId, name, dept) VALUES (3, 'Eve', 'IT');
cqlsh:labdemo> SELECT * FROM faculty;
```

```
 facultyid | dept | name
-----------+------+-------
         1 | Comp | Clark
         2 |   IT |  Dave
         3 |   IT |   Eve

(3 rows)
cqlsh:labdemo>
```

Conclusion: **study open-source database management tools like Cassandra, MySql, MariaDB and understand the steps for contributing in Cassandra.**

Debugging Cassandra Using Eclipse

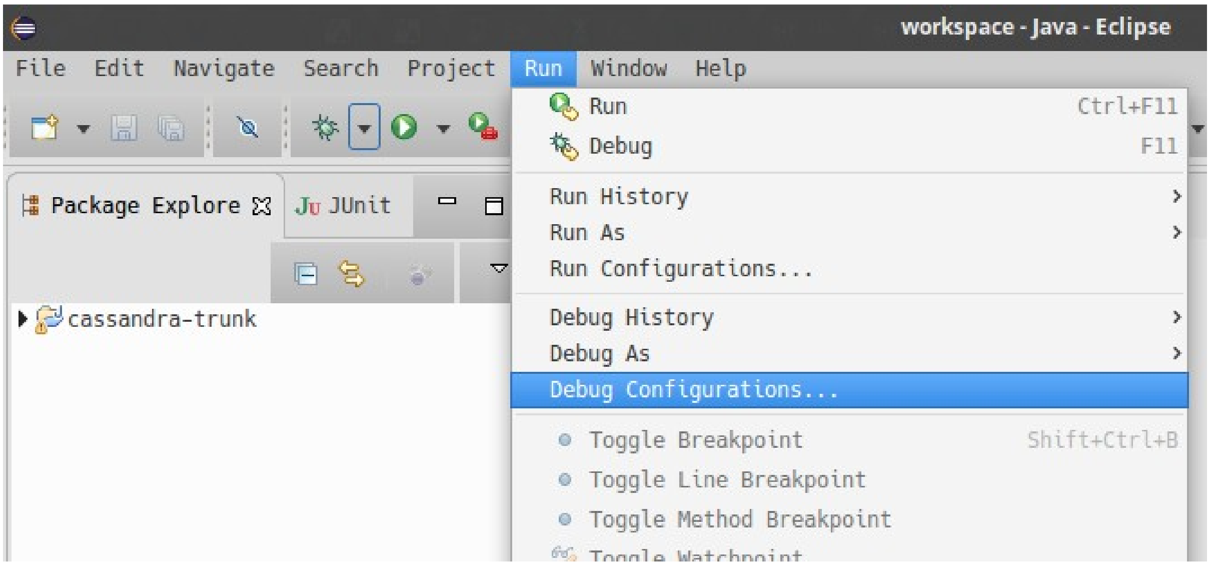
There are two ways to start a local Cassandra instance with Eclipse for debugging. You can either start Cassandra from the command line or from within Eclipse.

Debugging Cassandra started at command line

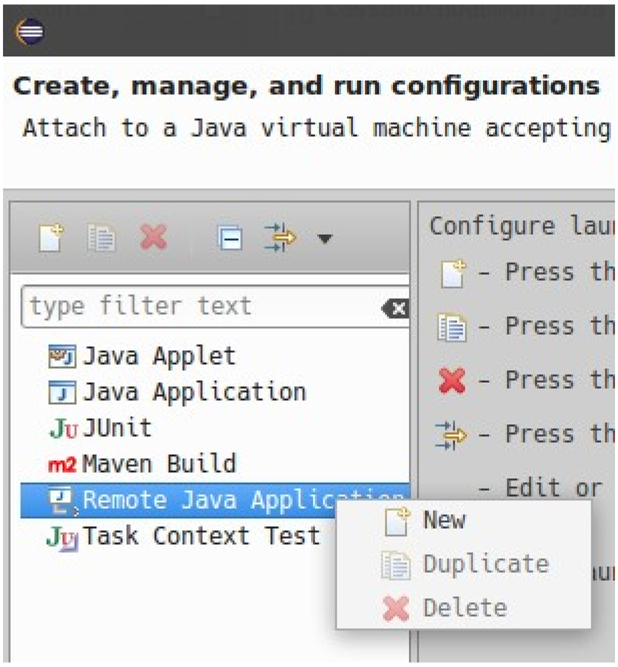
1. Set environment variable to define remote debugging options for the JVM: export JVM_EXTRA_OPTS="-agentlib:jdwp=transport=dt_socket,server=y,suspend=n,address=1414"
2. Start Cassandra by executing the ./bin/cassandra

Next, connect to the running Cassandra process by:

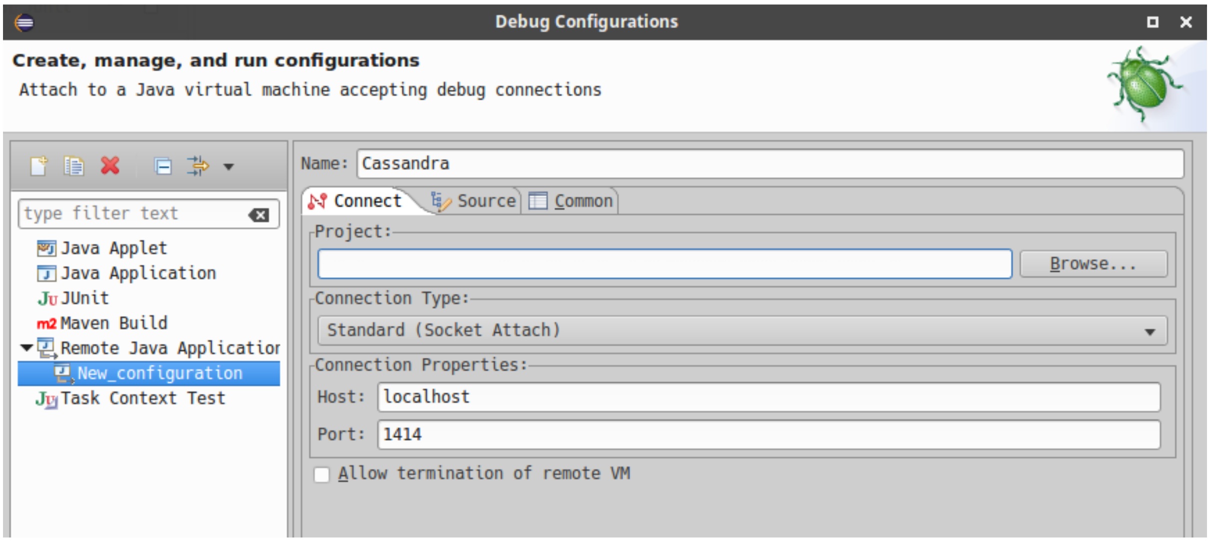
3. In Eclipse, select Run > Debug Configurations.



4. Create new remote application.



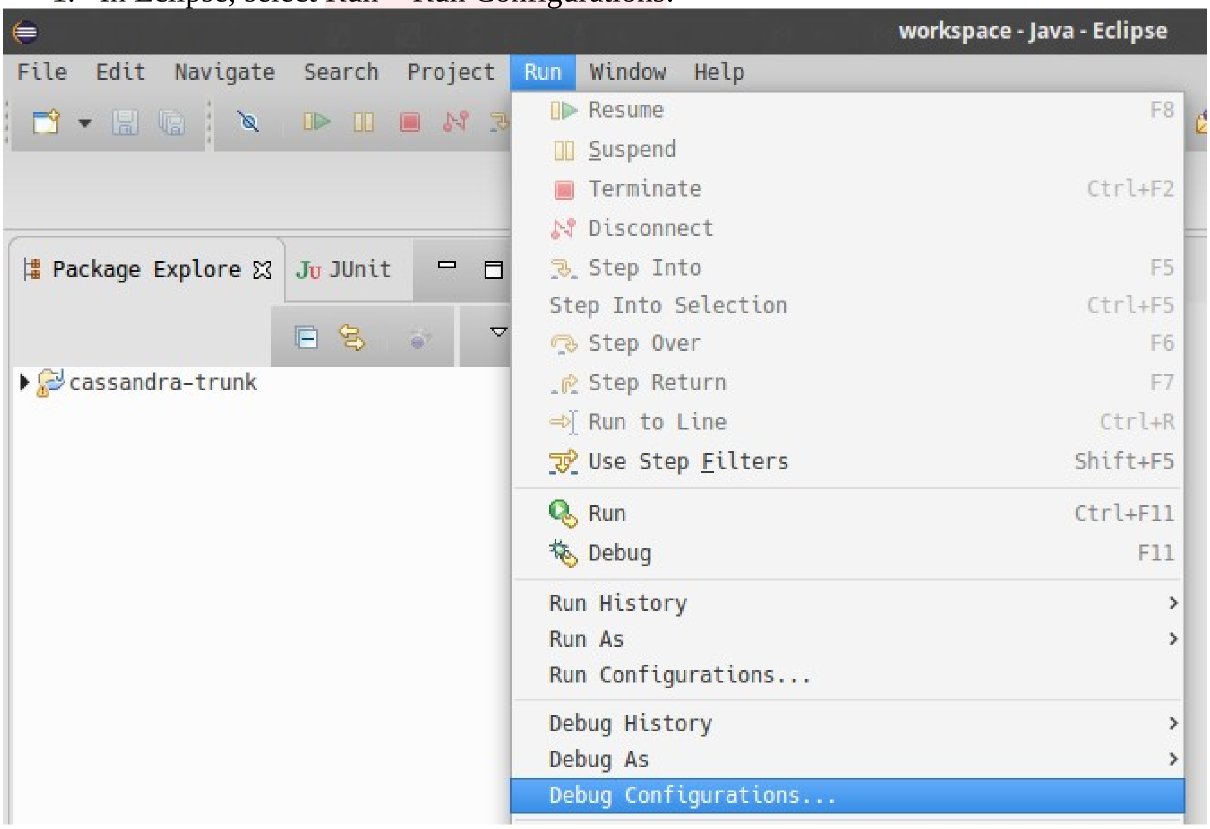
5. Configure connection settings by specifying a name and port 1414. Confirm Debug and start debugging.



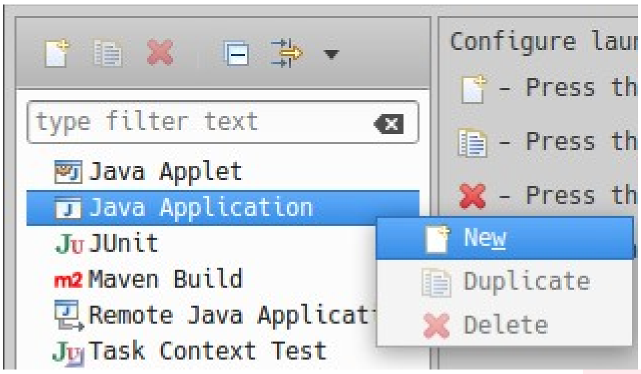
Debugging Cassandra started from Eclipse

Cassandra can also be started directly from Eclipse if you don’t want to use the command line.

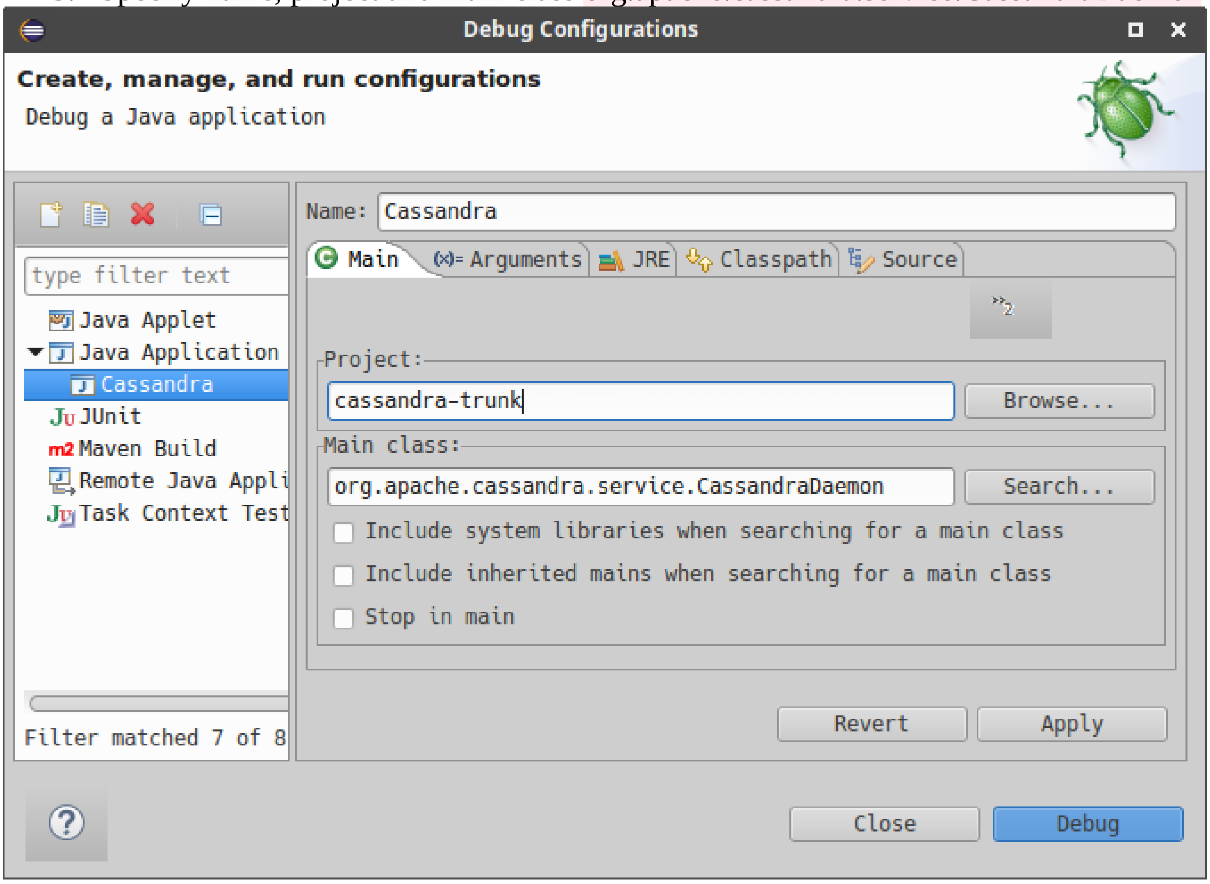
1. In Eclipse, select Run > Run Configurations.



2. Create new application.

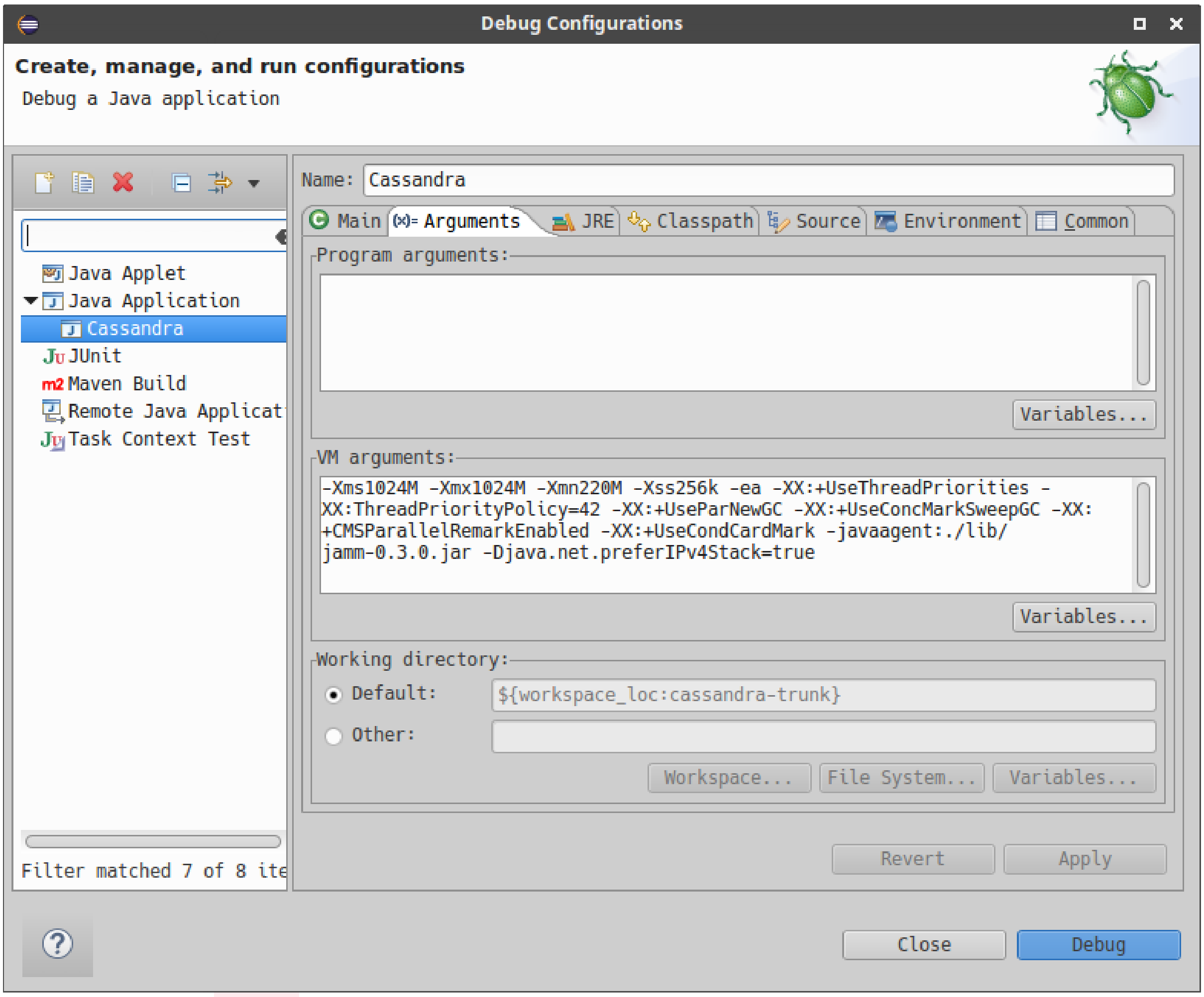


3. Specify name, project and main class org.apache.cassandra.service.CassandraDaemon



4. Configure additional JVM specific parameters that will start Cassandra with some of the settings created by the regular startup script. Change heap related values as needed.

```
-Xms1024M -Xmx1024M -Xmn220M -Xss256k -ea -XX:+UseThreadPriorities -XX:ThreadPriorityPolicy=42 -XX:+UseParNewGC -XX:+UseConcMarkSweepGC -XX:+CMSParallelRemarkEnabled -XX:+UseCondCardMark -javaagent:./lib/jamm-0.3.0.jar -Djava.net.preferIPv4Stack=true
```



5. Confirm Debug and you should see the output of Cassandra start up in the Eclipse console. You can now set breakpoints and start debugging!

General notes

You may sometimes encounter some odd build failures when running the ant commands above. If you do, start ant with the realclean option:

```
ant realclean
```

Remember that all the tasks mentioned above may depend on building source files. If there are actual compilation errors in the code, you may not be able to generate project files for IntelliJ Idea, Netbeans, or Eclipse. It is especially important that you have imported the project adequately into IDE before doing merges or rebases. Otherwise, if there are conflicts, the project cannot be opened in IDE, and you will be unable to use any fancy conflict resolution tools offered by those IDEs.

Sample Table Creation and Query Firing in Cassandra using Docker Image

1. Start Cassandra Container
docker run --name my-cassandra -p 9042:9042 -d cassandra:latest

if getting error message for permission denial use
sudo docker run --name my-cassandra -p 9042:9042 -d cassandra:latest

2. Connect to Cassandra with CQLSH
docker exec -it my-cassandra cqlsh

3. Create a Keyspace
```
CREATE KEYSPACE labdemo
WITH replication = {'class': 'SimpleStrategy', 'replication_factor' : 1};

USE labdemo;
```

4. Create a Sample Table
```
CREATE TABLE faculty (
   facultyId int PRIMARY KEY,
   name text,
   dept text
);
```

Step 4: Insert Records
```
INSERT INTO faculty (facultyId, name, dept) VALUES (1, 'Clark', 'Comp');
INSERT INTO faculty (facultyId, name, dept) VALUES (2, 'Dave', 'IT');
INSERT INTO faculty (facultyId, name, dept) VALUES (3, 'Eve', 'IT');
```

Step 5: Query the Table
```
SELECT * FROM faculty;
```

Expected Output:
```
 facultyId | name  | dept
-----------+-------+--------
         1 | Clark | Comp
         2 | Dave  | IT
         3 | Eve   | IT
```

```
cqlsh:labdemo> CREATE TABLE faculty (
    facultyId int PRIMARY KEY,
    name text,
    dept text
);
cqlsh:labdemo> INSERT INTO faculty (facultyId, name, dept) VALUES (1, 'Clark', 'Comp');
INSERT INTO faculty (facultyId, name, dept) VALUES (2, 'Dave', 'IT');
INSERT INTO faculty (facultyId, name, dept) VALUES (3, 'Eve', 'IT');
cqlsh:labdemo> DESCRIBE TABLES;

faculty  students

cqlsh:labdemo> DESCRIBE KEYSPACES;

labdemo  system_auth         system_schema  system_views
system   system_distributed  system_traces  system_virtual_schema

cqlsh:labdemo> SELECT * FROM faculty;

 facultyid | dept | name
-----------+------+-------
         1 | Comp | Clark
         2 |   IT |  Dave
         3 |   IT |   Eve
```